

Child Labor and Sibling Schooling: Evidence from 1920 America

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Abstract

If an older child's earnings relax a family's budget constraint, this may allow younger siblings to remain in school. We revisit this mechanism using state minimum-working-age laws and the full-count 1920 US Census. We compare children of the same age that also have older siblings of identical ages across states with different child labor laws. Exposure to more lax child labor laws increased older-sibling work, but we find little evidence of a large increase in the school attendance of their younger siblings. Our preferred estimate is 1.7 percentage points, with a state-clustered 95 percent confidence interval ranging from -1.9 to 5.3 points, which rules out the 6.6-point increase implied by earlier household-level evidence. With this in mind, we reproduce this evidence in both the original one-percent sample and the full count and show that, in fact, this effect is wrongly interpreted as a spillover. Finally, we link families to the 1910 Census and find a near-zero school attendance response among vulnerable families with low paternal occupational standing or an illiterate parent.

Keywords: child labor, school attendance, intrahousehold allocation, historical census

JEL codes: J13, J22, I21, N31

1 Introduction

Child labor laws may not only affect the children who become more or less likely to work, but may also change resource allocation within their families. For instance, a central implication of economic models of child labor is that an increase in the hours worked of an older child may help finance a younger child’s schooling. This mechanism is also consistent with historical accounts of the family economy (Basu and Van, 1998; Baland and Robinson, 2000; Parsons and Goldin, 1989) as well as modern experimental evidence, which has shown that the income received by one member of a household can alter other members’ work and education choices (Rhodes et al., 2026). We explore the specific question of whether an older child’s earnings increase the amount of schooling that younger siblings acquire. Identifying this effect is difficult, however, because the conditions that influence the likelihood that a child works also influence the likelihood that their siblings go to school. At the same time, the answer to this question is crucial for understanding the overall incidence of child-labor policy.

We study this question using state minimum-working-age laws combined with the full-count 1920 US Census. Because the full count is so large, we are able to compare children with exactly the same roster of sibling ages, which we find to be an important control. For each younger child, we measure the work status and exposure to child labor laws among their older siblings aged 10–16. Our preferred specification compares younger children of the same age who have the same number of older siblings who are exactly the same ages. In other words, their sibling age structures are identical, but differences in state law induce different amounts of work among those older siblings.

In short, exposure to more lax child labor law strongly increases older-sibling employment, but we find no statistically significant increase in younger siblings’ school attendance. In our sample of 1.76 million children, the school attendance response is equal to 1.7 percentage points, with a 95 percent confidence interval that ranges from -1.9 to 5.3 points. When we restrict the sample to the two most common working age thresholds of 12 and 14, the estimates range from -2.9 to 1.7 points. None of the specifications provides evidence of a school attendance gain arising from an increase in older-sibling work.

We then ask whether there is a school attendance gain among the families most likely to depend on their children’s earnings. To do this, we link families to the 1910 Census and exploit measures of paternal occupational standing and parental literacy in that Census (before the 1920 work and schooling decisions). Again, exposure to a more lax child labor law strongly predicts older-sibling work within these groups, however the school attendance estimates remain near zero, even among families with low paternal occupational standing or an illiterate parent.

Our findings differ from the existing evidence. In particular, Manacorda (2006) instruments the share of children in a family who work with their exposure to child labor law thresholds and finds an increase in school attendance among children in that family. For a family with two children aged 10–16, the estimate implies that moving one child into work increased the other’s school attendance by 6.55 percentage points. Our preferred confidence interval rules out an effect of that magnitude. We show that the difference in estimates reflects the way in which the treatment and instrument

are defined. In the “household” approach ([Manacorda \(2006\)](#)), both variables include the child whose schooling response is being estimated.¹ We show why this distinction matters by reproducing the household estimate in both the original one-percent Census sample and the full count. We show that families with only one child aged 10–16 receive 27 percent of the two-stage least-squares weight and contribute 89 percent of the school attendance coefficient. Clearly there is no scope for spillovers in these families and excluding these observations eliminates the large positive estimate.

This paper connects research on child-labor policy with evidence on household resource allocation and sibling spillovers. Existing research on compulsory-schooling and child-labor laws asks how legal restrictions affected the children directly covered by the law ([Moehling, 1999](#); [Lleras-Muney, 2002](#); [Clay et al., 2021](#)). [Janas \(2026\)](#) finds that reduced youth employment opportunities during the Great Depression increased educational attainment primarily among boys from higher-income families, highlighting the importance of household resources. We ask whether older siblings’ work provided resources that increased younger children’s school attendance. Existing evidence on household resource allocation provides a reason to expect effects on other children as well. In South Africa, [Edmonds \(2006\)](#) finds that variation in pension eligibility among elderly men increased school attendance for children in their household while also reducing the children’s hours of work. Recent related work on sibling spillovers shows that shocks to one child can affect other children in the family ([Black et al., 2021](#)). Overall, these findings suggest that a sibling financing mechanism is plausible, however they fall short of demonstrating that when an older child works more, their younger siblings are more likely to attend school.

The remainder of the paper is organized as follows. [Section 2](#) describes the historical setting. [Section 3](#) introduces the data and samples. [Section 4](#) introduces the empirical design and [Section 5](#) presents the results. [Section 6](#) examines what the household-average estimate identifies and [Section 7](#) extends our analysis to focus on economically vulnerable families. [Section 8](#) concludes.

2 Child Labor and Schooling in 1920

In the early twentieth century, two large and related changes were reshaping childhood in the United States. First, child labor had been declining from its late nineteenth-century peak while, at the same time, the high school movement was increasing access to secondary education. Nevertheless, adolescents still commonly worked, especially within farm households, and continued schooling beyond elementary school was not guaranteed. Children in their early teens could attend school, work, combine the two, or do neither. How to use an adolescent’s time was therefore an important household decision ([Goldin and Katz, 2008, 2011](#)).

Children’s work could be particularly important to households with limited access to credit or insurance against income shocks. This decision has historically been framed as a household allocation problem rather than a choice made by the child ([Parsons and Goldin, 1989](#)). In this

¹The specification does control for the child’s own exposure to the law, but it assumes that the law has the same direct effect on attendance regardless of family size. If the effect differs across family sizes, the model can mistake part of the child’s own response for a sibling effect, which is what we find.

context, a binding household resource constraint can make child labor privately necessary, even when the return to schooling is large (Basu and Van, 1998; Baland and Robinson, 2000; Edmonds, 2007). And if an older child’s earnings relax the household constraint, the family can in principle use the additional resources to keep a younger child in school. This is the specific household response that we test.

The federal government had also begun to regulate child labor by 1920, though we do not exploit national-level variation. Specifically, Title XII of the Revenue Act of 1918 taxed firms that employed children below specified ages,² though it excluded the bulk of agricultural and unpaid family work. State work-permit and compulsory-attendance laws therefore continued to determine the age at which market employment became legally available and this is the cross-state variation that we exploit. Our outcomes include household-farm work and mutually exclusive combinations of work and school.

During this period there was substantial heterogeneity across states in child labor laws. At the same time, age of school-entry, compulsory-attendance laws, and work-permit rules often overlapped while enforcement varied across states. This institutional complexity may help explain why historical studies disagree about important state policies were to the decline of child labor. For instance, Moehling (1999) finds a limited role for early child-labor laws, whereas Lleras-Muney (2002) and Clay et al. (2021) find that compulsory-attendance and child-labor laws increased education. Research on more recent child-labor regulations also shows that households may respond along margins not directly targeted by the law (Bharadwaj et al., 2020). However, these studies primarily establish effects on the children directly exposed to regulation.

Following Manacorda (2006), we use the state minimum age for a work permit, a dataset originally assembled by Lleras-Muney (2002). Most states fall into two groups, with seven states allowing work permits at age 12 while 32 set the minimum at age 14. Four states used age 15, one used age 13, and four had a lower or unrecorded minimum. New Mexico has no recorded minimum and is coded as zero, Louisiana and Wyoming are coded as age 7, and Florida as age 8. These differences generate the first-stage variation in the empirical design. Appendix Table 8 reports the details for each state.

For a child of age a in state s , legal exposure to the possibility of work is

$$E_{is} = \max\{a_i - L_s + 1, 0\}, \quad (1)$$

where L_s is the minimum work age in state s . We set legal exposure to zero below the minimum age and it then increases by one in each year beyond that. The measure therefore captures how many years that work has been legally available to an older child. In our preferred specification, state-by-younger-child-age fixed effects absorb the younger child’s own legal exposure, which is

²Approved in February 1919, Title XII imposed a tax equal to 10 percent of net profits on mines and quarries employing children under age 16 and on factories, mills, workshops, and canneries employing children under age 14 or employing children aged 14–16 beyond specified hours (United States Congress, 1919). The tax was in force during the 1920 census reference period and was struck down in 1922 in *Bailey v. Drexel Furniture Co.* (U.S. Supreme Court, 1922).

determined by state law and age.

In the context of the household resource allocation model, this design yields a set of straightforward predictions. First, legal access to employment should increase work among older children. That work should then increase school attendance among their younger siblings, particularly where limited household resources would otherwise prevent attendance.

3 Data and Samples

3.1 The 1920 Census

Our main sample is drawn from the complete 1920 Census (Ruggles et al., 2025). The existing household evidence in Manacorda (2006) instead uses the one-percent IPUMS sample. By using the full count data we increase precision while, more importantly, these data allow us to compare younger children who have siblings of exactly the same ages and to measure work and legal exposure only among those older siblings. In contrast, the one-percent sample is not large enough to make these comparisons.

To maintain comparability with the existing evidence, we adopt the sample restrictions applied in Manacorda (2006). The sample consists of children aged 10–16 who lived with both parents in families containing only the parents and children aged 0–16. Mothers must be between 25 and 41 years old and fathers between 25 and 66. Appendix Section D.3 relaxes these restrictions while still requiring two parents in the household.

Market work is defined using the harmonized 1950 occupation code and excludes the household-farm categories. The outcome labeled “Any work” in Table 1 includes these categories. The Census reports whether a person attended school, college, or any educational institution between September 1, 1919 and January 1, 1920. As a result, the measure reflects recent school attendance. In addition, work is derived from the existence of a reported occupation, so work and attendance can overlap. Children are classified as fully idle when neither work nor attendance is recorded.

The resulting sample consists of 4,428,417 children in 2,652,969 families. School attendance was 88.7 percent, 8.7 percent of children reported some work, and 7.1 percent were neither working nor in school. The average work share among children aged 10–16 was 8.5 percent. Appendix Table 6 reports the distribution of the sample across family sizes.

We also reconstruct the household-level sample that is used in Manacorda (2006) from the one-percent Census sample. Applying the published restrictions and variable definitions produces 44,451 children in 26,645 families, compared with 44,402 children in 26,612 families reported by the author. The difference of 49 children has no effect on the estimates. We also draw a one-percent sample from our full-count for comparison, which contains 44,448 children in 26,635 families. We note that the estimation code that was deposited for the paper does not replicate the published tables, but rather applies different sample restrictions, a different definition of work, and a single binary instrument. Appendix A and Table 4 document the differences between the deposited code and the sample reported in the paper alongside our full-count implementation.

3.2 Linked 1910 variables

For the heterogeneity analysis in Section 7, we measure family characteristics ten years prior to the 1920 outcomes. Census Tree links allow us to connect children and their parents to the 1910 Census (Buckles et al., 2025). This linked sample contains 1,217,941 children in 846,145 families.

These linked families are not quite representative of the full 1920 sample and we take this into account in our heterogeneity analysis (the main specifications use the full count). For instance, the linked families are more likely to live on a farm and to have children in school, and they are less likely to have an illiterate father. As a robustness check, we re-weight the linked families to match the composition of the full-count 1920 sample. Here, the sample weights balance state, sampled-child age structure, the number of younger children, sex composition, urban and farm status, race, parental ages, literacy, immigrant status, and housing tenure. This addresses selection on these observed characteristics, though not on unobserved factors that are correlated with the likelihood of matching across both censuses. Appendix Section E describes this approach in more detail and reports the reweighted estimates (which are qualitatively similar to the unweighted estimates).

3.3 Linked 1940 outcomes

We also link the younger siblings to the 1940 Census to estimate impacts on completed education. We retain links that maintain the same reported sex and reflect an age increase of 20 years. Since the likelihood of linking is more common for boys, our analysis focuses on sons only. Appendix Section D.4 reports summary statistics along with estimates where the sample is re-weighted to reflect the composition of eligible 1920 sons.

4 Empirical Design

Our main design follows the proposed flow of resources from older to younger children. Let i index a focal child in family f and state s , and let Y_{ifs} denote that child's outcome. Let $j = 1, \dots, N_{if}^O$ index child i 's older siblings aged 10–16, where N_{if}^O is the number of such siblings. The sample of primary interest is restricted to children with $N_{if}^O \geq 1$. For each sibling j , W_{jf} equals one if the census records market work and zero otherwise. Legal exposure, E_{js} , is defined as in (1) above. We define the corresponding older-sibling averages as

$$\overline{W}_{if}^O = \frac{1}{N_{if}^O} \sum_{j=1}^{N_{if}^O} W_{jf}, \quad \overline{E}_{if}^O = \frac{1}{N_{if}^O} \sum_{j=1}^{N_{if}^O} E_{js}. \quad (2)$$

Thus, $\overline{W}_{if}^O$ is the share of older siblings who work, while $\overline{E}_{if}^O$ is their average number of years of legal exposure. The preferred estimating equation is

$$Y_{ifs} = \beta \overline{W}_{if}^O + X'_{ifs} \gamma + \lambda_{sa_i} + \mu_{R_{if}^O} + \varepsilon_{ifs}, \quad (3)$$

where R_{if}^O is the complete roster of older-sibling ages, meaning their exact age combination. The terms λ_{sa_i} and $\mu_{R_{if}^O}$ denote state-by-focal-age and exact-roster fixed effects, respectively. The excluded instruments are

$$Z_{ifr}^O = \overline{E}_{if}^O \mathbf{1}\{L_s = r\}, \quad r \in \{0, 7, 8, 12, 13, 14, 15\}. \quad (4)$$

Here $\mathbf{1}\{L_s = r\}$ indicates that the minimum work age in state s is equal to r . For example, in a state with a minimum work age of 14, only $Z_{if,14}^O$ will be nonzero. Interacting exposure to the possibility of work with the seven different policy regimes allows the relationship between exposure and work to vary across states. The first stage therefore asks whether an older child with the a given set of older-sibling ages worked more under a state law that allowed for more years of legal access to employment.

The preferred specification includes state-by-(younger child)-age fixed effects along with fixed effects for the complete set of older-sibling ages. It therefore compares younger children of the same age whose older siblings are also exactly the same ages, but whose state laws produce different legal exposure. In this case, identification requires that, absent differences in child-labor laws, the relationship between older-sibling ages and younger-child school attendance would not vary across states in ways that are correlated with the legal working-age thresholds. We test this assumption by looking at school attendance patterns around each threshold and by allowing the relationship between older-sibling age and younger-child school attendance to vary flexibly across states.

Importantly, we also control for the exact age profile of siblings in order to avoid attributing to sibling work the ordinary fact that a 16-year-old sibling is both more likely to work and also more likely to be observed in a household at a later stage of family life relative to an 11-year-old. Other controls include the child's sex and birth order, the number and ages of younger children, family composition, urban and farm status, parental characteristics, housing tenure, race, and family size. Standard errors are clustered at the state level (the level at which the laws vary). We also report a wild-cluster bootstrap, leave-one-state-out estimates, and Anderson–Rubin tests.

Finally, we interpret the instrumental-variables estimate as the broad effect of law-induced older-sibling employment rather than as the effect of household income alone. This is in part because an older child's employment may alter the distribution of caregiving within the household or parental labor supply.

5 Older-Sibling Work and Younger Children's Schooling

The first stage is reported in Appendix Table 9 and indicates that exposure to a lower child labor threshold strongly predicts older-sibling work. Table 1 then reports the effect of older-sibling work on younger siblings. Columns (1) and (2) report school attendance with and without fixed effects for a given older-sibling age combination. Column (2) is our preferred specification and indicates a statistically insignificant point estimate of 0.017. Columns (3) and (4) use the same preferred specification to test for an effect on younger-child work and for an effect on neither working nor

attending school. Appendix Tables 9 and 11 report several robustness specifications.

Table 1: Does Work by Older Siblings Affect Younger Children?

Younger child's outcome	School attendance		Any work	Neither
	(1)	(2)	(3)	(4)
Older-sibling market-work share	−0.025*** (0.006)	0.017 (0.018)	0.013 (0.016)	−0.008 (0.014)
Older-sibling age-combination FE	No	Yes	Yes	Yes
Outcome mean	0.927	0.927	0.057	0.057
Observations	1,756,540	1,756,540	1,756,540	1,756,540

Notes: 2SLS estimates for younger children. *Any work* includes household-farm work. *Neither* means neither work nor school attendance. The treatment covers older siblings aged 10–16, instrumented by their mean legal exposure interacted with the seven legal regimes. All columns include state-by-younger-child-age fixed effects and the controls in Section 4. Standard errors in parentheses are clustered by 48 states. First stages appear in Appendix Table 9. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

For comparison, the Manacorda (2006) household-level benchmark relates each child's school attendance to the *share of all children* aged 10–16 in the family who work, using instruments based on their *average* exposure to a child-labor law. In contrast, using our full sample we are able to focus on the treatment and outcome of specific children with a full roster of sibling controls. Section 6 examines the implications of that distinction.

For the schooling impact reported in column (2), the state-clustered standard error is 0.018, implying a 95 percent confidence interval ranging from −0.019 to 0.053. This interval includes zero but, notably, not the 0.0655 household benchmark reported by Manacorda (2006). A wild-cluster Anderson–Rubin test also does not reject zero ($p = 0.759$) but rejects the household benchmark ($p = 0.006$). In addition, the leave-one-state-out estimates range from 0.006 to 0.023. Overall, the results reject the effect size reported in Manacorda (2006) but do not rule out smaller impacts.

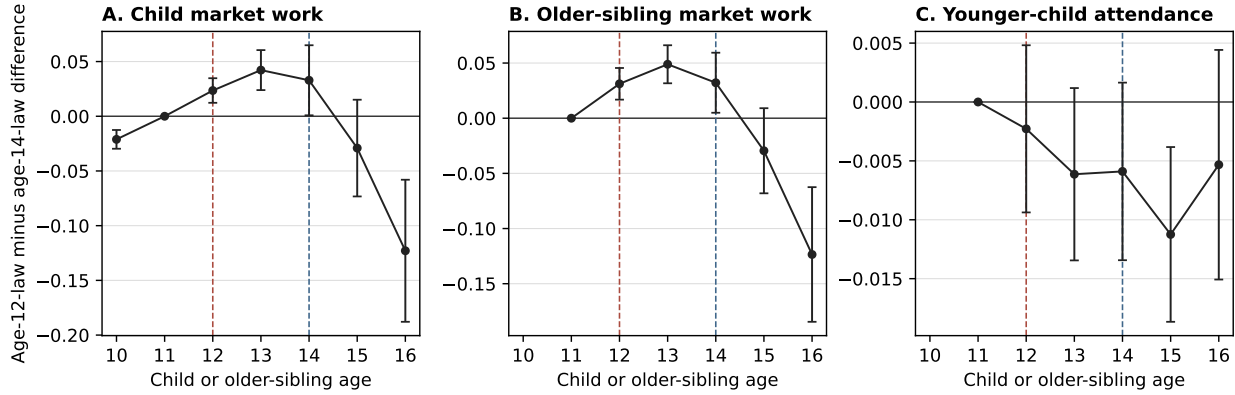
Appendix Table 9 reports law-specific first stages and reduced forms alongside each policy regime's contribution to the pooled estimate. We find that five of the seven school attendance reduced form estimates are negative and the positive coefficients are not statistically significant (Appendix B).³

The age-12 observations are responsible for 61 percent of the observation-group weight and contribute 1.22 percentage points to the 1.72-point pooled estimate. The two age-7 states contribute another 0.81 points despite receiving only 4.5 percent of the weight. Observations in the most common regime, age 14, cover 32 states and receive 20 percent of the weight but contribute essentially nothing (−0.009 percentage points) to the pooled estimate. Panel B of the same table shows that excluding the unrecorded, the age-7, and the age-8 regimes leads to an estimate of 0.011 (0.017). Restricting the sample to the 39 states with minimum work ages of 12 or 14 gives 0.012 (0.018). As

³The asymptotic Anderson–Rubin confidence set is empty, indicating that a common effect fits the law-specific moments poorly. The wild-cluster test does not reject zero.

a result, we conclude that the rejection of the [Manacorda \(2006\)](#) benchmark does not depend on the low-age regimes, even though they contribute substantially to the preferred point estimate.

Allowing the relationship between older-sibling age and school attendance to differ flexibly across states helps to test the sensitivity of our preferred specification. We find that state-specific linear trends explain 37.5 percent of the baseline variance and, when added to the preferred specification, lead to a school attendance coefficient of -0.029 (0.018). The rejection of the benchmark household-level estimate therefore survives this relaxation.⁴ Appendix Section D.1 reports the identifying variation and inference for each specification.



Coefficients are relative to the age-11 difference. Dashed lines mark the two legal thresholds.

Figure 1: Work and attendance profiles under the age-12 and age-14 laws

Notes: The figure compares states with minimum work ages of 12 and 14. Each point is the age-12-law minus age-14-law difference relative to the difference at age 11. Bars are 95 percent confidence intervals with state-clustered standard errors. Panel A uses all children and plots their own market work by age. Panels B and C use the sample of younger children that have exactly one older sibling and plot the older sibling's work and the younger child's school attendance as a function of the older sibling's age. Dashed lines mark the two legal thresholds.

Figure 1 makes the comparison of effects across legal regimes visible. Relative to states with a minimum working age of 14, work status increases among children in age-12 states when they reach ages 12 and 13, and the same pattern appears for older siblings. However, younger children's school attendance does not increase when this additional older-sibling work appears. In fact, its relative changes are negative for ages 12 through 16 and statistically different from zero only at age 15. The figure therefore shows a strong first stage without the concurrent attendance response implied by a sibling financing model.

The first two columns of Table 1 show why it is important to hold the ages of older-siblings fixed. Column (1) includes state-by-younger-child-age fixed effects but does not include the exact older-sibling roster. This specification produces a school attendance coefficient of -0.025 . However, this specification, in part, compares younger children whose older siblings are at different stages of the work-school transition. Holding the exact older-sibling age combination fixed in column (2) produces an imprecise estimate of 0.017.

⁴Quadratic controls explain only 3.0 percent of the variance and lead to an estimate of 0.232 (0.158). Using the common age-12 and age-14 law sample, the quadratic estimate is 0.293 (0.187) and the remaining first stage is weak.

The preferred estimates for younger-child work and the broader idleness measure are also small and imprecise. Columns (3) and (4) of Table 1 suggest there is little adjustment among younger siblings along alternative margins.

As a further test of the relationship, we reverse the direction of the effect and use younger-sibling work to explain the outcomes of older children. These estimates are sensitive to age composition and the attendance coefficient falls from an implausible -1.231 with state fixed effects to -0.168 with a standard error of 0.117 when the exact younger-sibling age roster is held fixed. We report these estimates in Appendix C.

Finally, we explore two additional approaches. First, we estimate a specification that compares neighboring counties across state borders, however this approach produces a weak first stage. Second, we estimate our primary specification on the linked sons’ 1940 sample. This produces a strong first stage but the second stage estimates are imprecise. Appendix Tables 12 and 14 report these results.

6 Comparison to the Household Estimate

Our preferred design measures work status and exposure to child labor laws among older siblings only—i.e., among the children who could potentially be exposed. The household specification in [Manacorda \(2006\)](#) instead averages across all children aged 10–16. Let $k = 1, \dots, N_f$ index these children in family f . The share of a family members who works is therefore

$$\overline{W}_f = \frac{1}{N_f} \sum_{k=1}^{N_f} W_{kf}. \quad (5)$$

[Manacorda \(2006\)](#) then instruments this share with the average exposure to child labor laws of these same children, interacted with indicators for the seven child-labor policy regimes. The regression also controls for the younger child’s own exposure to the laws, along with other controls and state fixed effects. Thus, both the treatment and the instruments include the younger child.

Panel A of Table 2 shows that this household estimate is reproducible. The [Manacorda \(2006\)](#) school attendance coefficient is 0.131 . Applying our specification to the data deposited with the paper produces an estimate of 0.133 , with the same standard error (after rounding). Applying this specification to our full-count data produces an estimate of 0.134 . We therefore conclude that the difference between the household results and our results is not due to sampling error or data construction.

In a family with two children aged 10–16, the household coefficient of 0.131 implies a 6.55-percentage-point school attendance gain from moving one child into work. However, the household coefficient is not, in fact, the estimate of interest. This is because the family work share, and the associated instruments, include the younger child. The issue is clearest when a family has only one child aged 10–16, in which case the household measure clearly only captures that child’s work and exposure to the law. [Manacorda \(2006\)](#) does in fact control for the focal child’s own exposure to the law; however, this does not fully solve the problem because the coefficient on the control is then

constrained to be the same across family sizes.

Figure 2 decomposes the full-count estimate by the number of children aged 10–16. The left panel reports each group’s share of the residualized first-stage covariance. The middle and right panels show each group’s contribution to the work and school attendance coefficients, respectively.

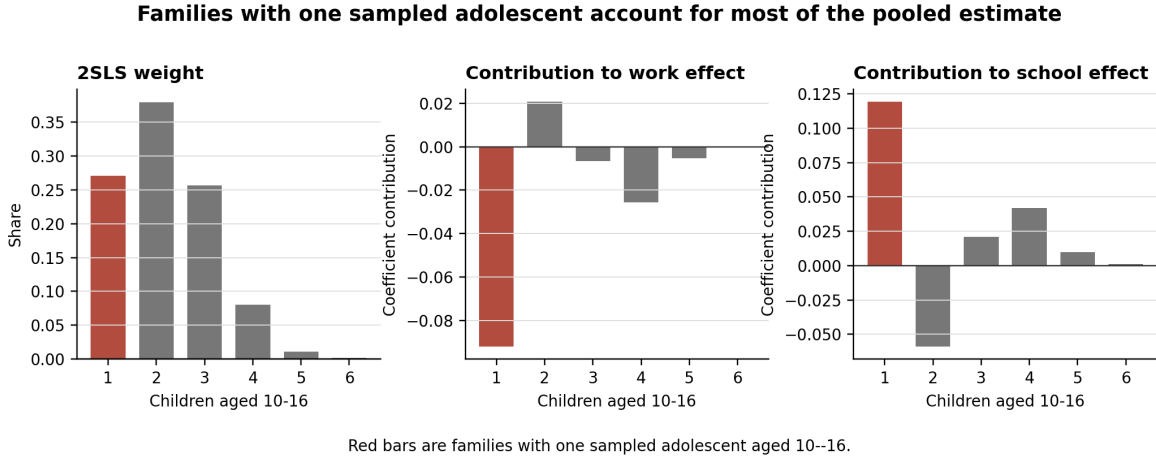


Figure 2: Family-size contributions to the household-average estimate

Notes: The figure decomposes the full-count 2SLS estimate by the number of adolescents aged 10–16 in each family. The treatment, instruments, and controls are residualized before constructing the group contributions (see Appendix B). Weights sum to one but need not be positive. Families contributing one sampled adolescent are shown in red.

We see that families with one sampled adolescent account for 1.40 million of the 4.43 million observations and receive 27.1 percent of the two-stage least-squares weight. Despite this modest weight, they generate 89 percent of the school attendance coefficient. Again, a family with only one sampled adolescent clearly cannot identify the spillover of interest. Moreover, families contributing two adolescents produce an effect in the opposite (“wrong”) direction.

With this in mind, panel B of Table 2 applies two corrections. First, it allows the focal (younger) child’s response to legal exposure to differ with the number of sampled adolescents. The attendance coefficient changes from 0.134 to -0.084 . In fact, simply excluding families with one sampled adolescent produces an estimate of -0.022 (0.019), as reported in Appendix Table 5. Second, the table removes the focal child from both the work measure and the instruments. However, it is important to note that the pooled leave-one-out estimate is negative because it compares children with siblings of different ages. Once the complete sibling-age roster is held fixed, the school attendance coefficient is 0.015 (0.026). And among families with two sampled adolescents, it is 0.012 (0.025).

The household coefficient reported by [Manacorda \(2006\)](#) is therefore reproducible and indeed reflects the response of a family to child-labor laws, but it does not isolate a transfer across siblings. Its positive school attendance coefficient is generated primarily by families in which the younger child also responds to the household-level treatment. Removing this child from the sample and holding the set of sibling ages fixed brings the school attendance estimate close to zero.

Table 2: From the Household Benchmark to Sibling Comparisons

Specification	Any work	In school	Neither
<i>Panel A. Household benchmark across samples</i>			
Published Table 6	−0.164** (0.075)	0.131* (0.076)	−0.025 (0.064)
Deposited microdata, published inference	−0.166** (0.075)	0.133* (0.076)	−0.026 (0.064)
Full count, state clusters	−0.109*** (0.030)	0.134*** (0.026)	−0.025* (0.014)
<i>Panel B. Separating the focal child from the treatment</i>			
Older child slopes by family size	0.058*** (0.020)	−0.084*** (0.016)	0.035*** (0.006)
Leave-one-out siblings, pooled	0.090*** (0.008)	−0.143*** (0.011)	0.055*** (0.008)
Leave-one-out, exact sibling-age roster	0.020 (0.013)	0.015 (0.026)	−0.014 (0.016)
Two sampled children, exact sibling age	0.048** (0.022)	0.012 (0.025)	−0.015 (0.017)
Observations, household average		4,428,417	
Observations, exact leave-one-out		3,029,079	

Notes: Panel A reports two-stage least-squares coefficients where the regressor is the family work share. The rows associated with the Manacorda (2006) paper use that article’s inference. The full-count estimates cluster by state. Panel B first allows the older child’s exposure response to vary with the number of sampled adolescents. Next, we remove that child from the treatment and instruments. Exact-age-roster controls absorb state-by-focal-age and sibling-age-roster fixed effects. Panel B standard errors are clustered by state. The household sample contains 4,428,417 observations while the exact leave-one-out sample contains 3,029,079. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Did Vulnerable Families Benefit?

It is possible that the small average estimate may conceal important gains for families that depended most heavily on their children’s earnings. As a proxy for this greater dependence, we first use the father’s occupation observed a decade earlier. Our primary measure ranks a father’s 1910 occupational-income score relative to other linked fathers in the same state. Families below the state median are then classified as having low paternal occupational standing. In addition, we explore the effect for families in which at least one parent was illiterate in 1910. While neither measure is a perfect reflection of family resources, they are intuitive measures of economic need that, importantly, predate our work and schooling outcomes by ten years and so cannot have been caused by the 1920 treatment.

Table 3 first shows that the main estimate using this linked sample is similar to the full count estimate above. Specifically, the school attendance coefficient is 0.008 with a standard error of 0.024, which is close to the full-count result. Splitting the sample by the father’s earlier occupational standing has little effect on the point estimates. The estimate is 0.012 for lower-standing families and 0.007 for higher-standing families, with state-clustered standard errors of 0.034 and 0.018, respectively. And again, exposure to child labor laws predicts older-sibling work in both groups (i.e., the first stage is strong).

Table 3: Family Background in 1910

1910 family background	Attendance effect	Attendance mean
All linked families	0.008 (0.024)	0.937
Low paternal occupation score	0.012 (0.034)	0.927
Higher paternal occupation score	0.007 (0.018)	0.952
At least one illiterate parent	0.001 (0.050)	0.853
No illiterate parent	−0.001 (0.020)	0.946
Paternal-occupation observations	1,217,941	
Parental-literacy observations	1,259,508	

Notes: Families are linked to their 1910 census characteristics. Low paternal occupation score means below the state median among linked families. Parental illiteracy predates the 1920 treatment and outcome. Each row uses the exact older-sibling-age-roster design from Table 1. The occupation and literacy samples contain 846,145 and 875,670 families, respectively. Standard errors are clustered by state. Stars use a t distribution with the number of states minus one degrees of freedom. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

The parental-literacy split indicates similarly small point estimates. The coefficient on school attendance is 0.001 when at least one parent was illiterate in 1910 and −0.001 when neither parent

was classified as illiterate. Exposure to a more lax child labor law predicts older-sibling work in both samples, but neither group shows a detectable school attendance gain. Restricting the analysis to the 39 states with minimum work ages of 12 or 14 also produces estimates near zero in both occupational groups.⁵

Appendix Table 15 presents estimates from the linked sample with weights applied to reproduce the composition of the full-count sample. Here, the school attendance estimate remains small and imprecise across the specifications.

A final possibility that we consider is that the broad sibling measure averages over children who are too far in age from the relevant schooling and work margins. With this in mind, Appendix Table 16 narrows the comparison to children aged 10–11 who have an older brother aged 12–14. In the full count, estimate is only statistically significant among families with low paternal occupational standing (-0.330 (0.167)), however the difference between this estimate and the estimate for high paternal standing (-0.255 (0.172)) is not statistically significant.

8 Conclusion

We revisit the literature that argues that work by one child within a family can help to finance a sibling’s schooling. We exploit state minimum-working-age laws and the full-count 1920 Census to separate children’s own responses to the law from effects operating across their siblings. The design compares younger children with identical older-sibling age structures across states where the laws induced different amounts of older-sibling work.

Our preferred school attendance estimate is 1.7 percentage points, with a state-clustered 95 percent confidence interval that ranges from -1.9 to 5.3 points. Across this specification and a range of alternative specifications the estimates range from -2.9 to 1.7 points. Importantly, all the estimates exclude the 6.55-point estimate in the related study from [Manacorda \(2006\)](#). We then further explore the potential for effects that are concentrated in vulnerable families or in later-outcome attainment. These specifications produce strong first stages (an effect on older child work) but produce imprecise outcome estimates in the second stage in each case.

Separately, we show that the household-level estimate in [Manacorda \(2006\)](#) cannot isolate a transfer between siblings. We do find that the positive school attendance effect appears in both the original paper’s one-percent sample (as reported in that paper) and in our full count. However, families with only one child generate 89 percent of the effect, which is clearly an error in the construction of the sample since these observations cannot identify a transfer between siblings.

Finally, it is important to note that these findings address a specific type of intra-household transfer, namely to finance schooling for a younger sibling. We do not address the broader importance of household resources. Unlike a cash transfer, an older child’s employment changes the family’s time allocation as well as its income. Additional earnings could be offset by lost caregiving or

⁵Appendix Table 10 reports heterogeneity using parental literacy recorded in 1920. Here the school attendance coefficient is -0.094 (0.019) among families with an illiterate parent, compared with -0.004 (0.047) among other families. This near-zero linked estimate therefore does not overturn the full-count result.

lost work by others within the household. Related evidence from the literature suggests why the form of the intervention may matter. In a randomized school transfer experiment run in Colombia, untreated siblings were found to attend less school when a different child received a schooling subsidy ([Barrera-Osorio et al., 2011](#)). Thus, the subsidy changed both the amount of household resources as well as the incentives to educate different children within the family. The Census data that we exploit record employment but not the resulting income gain, so we cannot separate an income effect from other household adjustments. As a result, the absence of a clear attendance benefit should therefore not be read as evidence that families did not share resources.

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A Benchmark Reconstruction and Sample Details

The archived estimation program for [Manacorda \(2006\)](#) uses an urban sample, different parent-age and occupation definitions, and one binary instrument rather than the seven instruments reported in the published article. With this in mind, we reconstruct the published empirical approach from the author’s deposited raw dataset using that article’s sample rules, along with its occupation definitions, child-labor-law-specific exposure instruments, controls, and clustering. Table 4 documents the sample differences and the main-text Table 2 reports the recovered coefficients.

Table 4: Sample and Code Comparisons

Sample or implementation	Children	Families	Main distinction
Published article	44,402	26,612	Counts reported in Table 6
Archived <code>reg.do</code> sample	54,391	34,118	Urban-only sample, binary eligibility instrument, and earlier variable definitions
Deposited cleaned data	42,478	25,780	Published restrictions once the archived cleaning program removes households with same-age children
Raw deposited extract	44,451	26,645	Our reconstruction of the published restrictions and seven-instrument design
Deterministic one-percent full count	44,448	26,635	Same construction as the full-count sample
Full-count Census	4,428,417	2,652,969	Publication-matched family restrictions applied to the complete Census

Notes: The close coefficient match in Table 2 shows that the small difference in samples do not explain differences in the results. Since the article abstract and the archived program code describe an urban sample, we recover the published sample counts by treating urban status as a control. We then report an urban-only sibling estimate separately.

Table 5: Household Reconstruction and Urban-Sample Checks

Specification	Attendance effect	Observations
Household model, all families	0.134*** (0.026)	4,428,417
Household model, at least two sampled adolescents	−0.022 (0.019)	3,029,079
Household model, urban households	0.195*** (0.017)	1,964,764
Older-sibling model, urban households	−0.017 (0.015)	711,873

Notes: The reported standard errors are clustered by state. The household models include the focal (younger) child in the measures of family work and legal exposure. The second row excludes families contributing only one adolescent aged 10–16. The final row then uses only older siblings in the treatment and instruments.

Table 5 reports two additional reconstruction checks. Excluding families with only one sampled adolescent eliminates the positive household estimate. Restricting the sample to urban households instead strengthens the household coefficient, but leaves the older-sibling (first stage) estimate near zero.

Table 6: Full-Count Sample Composition

Sampled adolescents aged 10–16	Child observations	Families	Share of children
1	1,399,338	1,399,338	0.316
2	1,660,800	830,400	0.375
3	1,006,572	335,524	0.227
4	311,812	77,953	0.070
5	43,920	8,784	0.010
6 or more	5,975	970	0.001
Total	4,428,417	2,652,969	1.000

Notes: Publication-matched full-count sample. Families are two-parent nuclear families satisfying the parent-age and child-age restrictions described in Section 3.

B Formal Two-Stage Least-Squares Decomposition

The household treatment can be written as $\bar{W}_f = N_f^{-1}W_{if} + (N_f - 1)N_f^{-1}\bar{W}_{-i,f}$, where W_{if} is a given “focal” child’s work and $\bar{W}_{-i,f}$ is the average work among the other children. To learn which family structures generate the two-stage least-squares estimate, we first partial out the controls and fixed effects. Let M_X be the residual-maker for those controls. Define $\tilde{Y} = M_X Y$, $\tilde{D} = M_X D$, and $\tilde{Z} = M_X Z$, where D is family work and Z contains the seven child-labor-law-specific family-exposure instruments. The fitted treatment is

$$\hat{D} = P_{\tilde{Z}}\tilde{D} = \tilde{Z}(\tilde{Z}'\tilde{Z})^{-1}\tilde{Z}'\tilde{D}. \quad (6)$$

With one endogenous treatment, the pooled two-stage least-squares coefficient can then be written as

$$\hat{\beta} = \frac{\hat{D}'\tilde{Y}}{\hat{D}'\tilde{D}}. \quad (7)$$

We can then partition the observations into groups g , such as the number of sampled older children in the family or by different states’ legal-age thresholds. We can define

$$\omega_g = \frac{\hat{D}'_g\tilde{D}_g}{\hat{D}'_g\tilde{D}_g}, \quad \hat{\beta}_g = \frac{\hat{D}'_g\tilde{Y}_g}{\hat{D}'_g\tilde{D}_g}. \quad (8)$$

It follows that

$$\hat{\beta} = \sum_g \omega_g \hat{\beta}_g. \quad (9)$$

The weights sum to one but can be negative after residualization. Each contribution is given by $\omega_g \hat{\beta}_g$. As Angrist (2014) similarly shows, group averages need not identify spillovers. In the case here, common own-exposure coefficients across family sizes can confuse own and sibling effects.

At the level of different child-labor regimes, shared roster effects and controls ultimately link observations across regimes. In other words, an instrument that is initially zero outside its legal regime can be nonzero there after residualization. After accounting for the controls, predicted work status in each group therefore reflects information from all seven instruments, not just the instrument associated with that group’s law. As a result, its contribution, $\hat{D}'_g \tilde{Y}_g / (\hat{D}' \tilde{D})$, can be positive even when its own instrument’s reduced-form coefficient is negative. Table 9 reports both objects where we see that group contributions sum to the pooled estimate, not to a weighted average of the reduced-form/first-stage ratios.

Table 7: State-Level and Weak-Instrument-Robust Diagnostics

Diagnostic	Null or estimate	<i>p</i> -value
First-stage partial R^2 , preferred model	0.00269	–
State-clustered Anderson–Rubin, $\beta = 0.0000$	64.789	0.000
State-clustered Anderson–Rubin, $\beta = 0.0655$	211.632	0.000
Wild-cluster Anderson–Rubin, $\beta = 0.0000$	66.168	0.759
Wild-cluster Anderson–Rubin, $\beta = 0.0655$	216.136	0.006
Leave-one-state-out range	[0.006, 0.023]	–
Pooled-exposure permutation: older-sibling work	-0.000	0.953
Pooled-exposure permutation: attendance	-0.001	0.383

Notes: Anderson–Rubin tests use the law-specific instruments jointly. The partial R^2 is from the preferred first stage after absorbing controls and fixed effects. The wild-cluster test uses null-imposed Rademacher weights at the state level. The permutation uses a separate pooled-exposure specification and reassigns the age-12 and age-14 laws across states while preserving their frequency. The benchmark $\beta = 0.0655$ is the attendance effect implied by one additional working child in a two-child family.

The asymptotic and wild-cluster Anderson–Rubin tests disagree about whether the effect is in fact zero. The asymptotic test rejects zero, while the null-imposed wild-cluster test does not. With 48 policy clusters and reduced forms that differ across legal regimes, we place greater weight on the wild-cluster result but note that the tests are inconclusive.

In the common-law states sample (those with age 12 or 14 thresholds), the pooled-exposure attendance estimate is 0.603 (0.843), compared with 0.012 (0.018) with separate instruments for the two regimes. The permutation diagnostic in Table 7 also uses a separate pooled-exposure specification, not the preferred seven-instrument model.

Table 8: State Child-Labor and Compulsory-Schooling Laws in 1920

State	Work age	Entry	Leaving	School exempt.	Work permit	Cont. school	Cont. age
Alabama	12	8	16	8	—	No	—
Arizona	14	8	16	8	5	No	—
Arkansas	14	7	15	8	4	No	—
California	14	8	15	8	8	Yes	16
Colorado	14	8	16	8	—	Yes	16
Connecticut	14	7	16	—	5	Yes	16
Delaware	14	7	14	—	—	No	—
Florida	8	8	14	—	—	No	—
Georgia	12	8	14	4	—	No	—
Idaho	15	7	16	8	8	No	—
Illinois	14	7	16	—	5	Yes	16
Indiana	14	7	16	—	5	Yes	16
Iowa	14	7	16	8	6	No	—
Kansas	14	8	16	8	8	No	—
Kentucky	14	7	16	—	5	No	—
Louisiana	7	7	14	8	—	No	—
Maine	14	7	15	—	5	No	—
Maryland	13	7	15	—	5	No	—
Massachusetts	14	7	16	7	4	Yes	—
Michigan	14	7	16	8	6	No	—
Minnesota	14	8	16	8	—	No	—
Mississippi	12	7	14	8	—	No	—
Missouri	14	8	16	—	—	No	—
Montana	14	8	16	—	—	No	—
Nebraska	14	7	16	8	8	Yes	16
Nevada	14	8	16	8	—	No	—
New Hampshire	14	8	16	8	5	Yes	—
New Jersey	14	7	16	—	5	No	—
New Mexico	—	7	14	—	—	No	—
New York	14	8	16	8	8	Yes	16
North Carolina	12	8	14	—	—	No	—
North Dakota	14	7	15	8	—	No	—
Ohio	15	8	16	6	6	Yes	16
Oklahoma	14	8	16	—	—	No	—
Oregon	14	9	15	8	—	No	—
Pennsylvania	14	8	16	—	6	Yes	16
Rhode Island	14	8	16	8	—	No	—
South Carolina	14	8	16	—	—	No	—
South Dakota	15	8	16	8	—	No	—
Tennessee	14	8	16	8	—	No	—
Texas	12	8	14	—	4	No	—
Utah	12	7	16	8	—	No	—
Vermont	15	8	16	8	6	No	—
Virginia	14	8	12	4	—	No	—
Washington	12	8	16	8	8	No	—
West Virginia	14	8	15	4	4	No	—
Wisconsin	14	7	16	8	5	Yes	17
Wyoming	7	7	14	—	—	No	—

Notes: Work age is the minimum legal employment age. Entry and leaving reflect the compulsory-schooling ages. School exemption and work permit report completed-schooling requirements. A dash denotes no recorded requirement. The continuation-school indicator includes evening and part-time schooling. Its 1920 values carry forward the 1915 coding and do not establish individual obligations during the census attendance window.

C Additional Sibling Results

Table 9 reports the law-specific first stages, reduced forms, and contributions to the pooled estimate, together with exercises that exclude the low-age states or that use only the common age-12 and age-14 laws. Table 10 reports full-count heterogeneity. The complete progression of older-to-younger specifications is available in the replication materials.

Table 9: Legal-Regime Coefficients and Contributions to the Pooled Estimate

<i>Panel A. Instrument coefficients and observation-group contributions</i>					
Minimum work age	States	First stage	Attendance reduced form	Observation-group weight	Contribution (pp)
Unrecorded	1	0.741 (0.610)	0.026 (0.116)	2.3%	0.004
7	2	2.749*** (0.619)	−0.519*** (0.126)	4.5%	0.813
8	1	3.198*** (0.614)	−0.143* (0.078)	1.4%	−0.137
12	7	2.162*** (0.629)	−0.326*** (0.097)	61.0%	1.218
13	1	9.227*** (0.656)	0.057 (0.088)	4.8%	0.220
14	32	7.557*** (1.370)	−0.243** (0.111)	20.0%	−0.009
15	4	8.214*** (1.594)	−0.046 (0.139)	6.0%	−0.387
Total	48			100.0%	1.722
<i>Panel B. Re-estimated attendance effects (percentage points)</i>					
Sample	States	2SLS estimate			
Exclude unrecorded, ages 7 and 8	44	1.052 (1.719)			
Common work ages 12 and 14	39	1.158 (1.764)			

Notes: All coefficients, standard errors, and contributions are in percentage points. Reported first stages and reduced forms are coefficients on the seven exposure instruments entered jointly. The last two columns instead partition observations by their state's law. Contributions sum to the pooled estimate in percentage points. Weights sum to one. Panel B refits the model on the indicated samples. All models include state-by-younger-child-age and exact older-sibling-age-roster fixed effects. Standard errors are clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Full-Count Heterogeneity in the Older-Sibling Estimates

Comparison	Group 0	Group 1	Difference
Rural / Urban	−0.126** (0.050)	−0.059** (0.028)	0.068*** (0.026)
Nonfarm / Farm	−0.021 (0.031)	−0.046 (0.067)	−0.025 (0.040)
Boys / Girls	0.007 (0.017)	0.023 (0.020)	0.016 (0.010)
Literate parents / At least one illiterate parent	−0.004 (0.047)	−0.094*** (0.019)	−0.090** (0.037)

Notes: Joint interacted two-stage least-squares models estimate both group coefficients and their difference in a common sample. Standard errors are clustered by state. Parental illiteracy is observed in 1920. The predetermined 1910 measures are reported separately. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Additional Identification and Scope Checks

D.1 State-specific age profiles

The exact-sibling-age design holds older-sibling ages fixed across states, but does not absorb state-specific differences in the extent to which those ages predict younger-child attendance. We examine this more demanding restriction by adding state-specific linear and quadratic terms in mean older-sibling age. Because the focal children are at least 10, their strictly older siblings are at least 11. Legal exposure under thresholds 0, 7, 8, and 12 is therefore linear in older-sibling age throughout this sample and the state-specific linear controls will absorb those four instruments.

Table 11: Identification and Age Profiles

Outcome	Exact roster	Linear profiles	Quadratic profiles	Quadratic ages 12 and 14
School attendance	0.017 (0.018)	−0.029 (0.018)	0.232 (0.158)	0.293 (0.187)
Observations	1,756,540	1,756,540	1,756,540	1,562,713
Families	1,242,863	1,242,863	1,242,863	1,104,895
States	48	48	48	39
Excluded instruments	7	3	3	1
First-stage Wald χ^2	104,891.3	235.4	34.6	7.6
Fitted-work variance (%)	100.0	37.5	3.0	2.1

Notes: The treatment is the share of strictly older siblings aged 10–16 in market work. All models retain state-by-focal-child-age and exact older-sibling-age-roster fixed effects. Linear and quadratic profiles are polynomials in mean older-sibling age interacted with state. Algebraically absorbed instruments are excluded before normalization. Fitted-work variance is expressed relative to the first column. Standard errors clustered by 1920 state are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The linear specification retains 37.5 percent of the fitted-work variance from the baseline

specification. Adding quadratic terms reduces this to 3.0 percent. The school attendance coefficient is 0.232, with a state-clustered 95 percent interval from -0.079 to 0.542 . Restricting the quadratic model to states with the common thresholds leaves only the age-14 instrument, with a first-stage Wald statistic of 7.6. The estimate in this case is 0.293, and the interval runs from -0.073 to 0.660 . We note that a wild-cluster Anderson–Rubin test rejects neither zero ($p = 0.178$) nor the 0.0655 benchmark ($p = 0.250$) in this common-state-law specification.

D.2 Neighboring county border design

As an alternative research design, we compare neighboring counties in states with minimum work ages of 12 or 14. In principle this is a strong research design, however the results are inconclusive and so we relegate discussion to this appendix. Historical 1920 boundaries identify 211 cross-state county pairs in 16 states where we have a sample of 102,733 children in 72,176 families. The first model absorbs county-pair-by-younger-child-age-by-older-sibling-roster and county-by-younger-child-age effects. The weights give each child equal total weight across pairs. Table 12 reports linear-profile and equal-pair-weight alternative specifications.

Table 12: School Attendance in the Border-County Comparison

Comparison	Attendance	Children	First-stage Wald χ^2	Wild AR p $\beta = 0$	Wild AR p $\beta = 0.0655$
National, common laws	0.012 (0.018)	1,562,713	35.644	0.0215	0.0027
Border counties	-0.297 (0.389)	106,924	2.842	0.6240	0.4857
Shared roster support	-0.217 (0.422)	102,733	2.086	0.7682	0.7143
Primary local comparison	-0.245 (0.938)	102,733	0.943	0.9599	0.9346
Local + linear profiles	-0.149 (0.838)	102,733	0.897	0.8550	0.7773
Equal county-pair weights	-0.762 (0.934)	102,733	2.946	0.6332	0.5913

Notes: The treatment is the share of older-siblings that work in a family. All comparisons use the age-12 and age-14 child labor laws. The first three rows use the national state-by-focal-child-age and exact older-sibling-roster controls, where only the sample changes. The local models absorb county-pair-by-focal-age-by-older-sibling-roster and county-by-focal-age effects. Inverse-membership weights give each child equal total weight except in the final row, which gives each county pair equal total weight. The national model has 39 state clusters. The border models have 16 state clusters. State-clustered standard errors are reported below the estimates. Wild-cluster Anderson–Rubin tests impose a common coefficient across instruments.

The first stage weakens even before adding the fixed effects. All border specifications are imprecise, and the wild-cluster Anderson–Rubin tests do not reject zero nor do they reject the

household benchmark. The comparison is therefore inconclusive as noted.

D.3 Broader family samples

We first allow both parents to be aged 18–99 and then admit adult children and extended-family structures. Siblings continue to share the same identified mother and father. All other elements of our primary specification are retained.

Table 13: School Attendance Under Broader Family Restrictions

Outcome	Original	Parental ages 18–99	Expanded families
School attendance	0.017 (0.018)	0.008 (0.020)	−0.013 (0.015)
Observations	1,756,540	2,274,394	4,876,679
Families	1,242,863	1,603,295	3,341,074
States	48	48	48

Notes: The treatment is the share of strictly older siblings aged 10–16 in market work. All models apply state-by-focal-child-age and exact older-sibling-age-roster fixed effects. The samples are nested and therefore require two identified co-resident parents. All three models also control for children over 16 and other household members. The second row relaxes parental age restrictions. The third also relaxes family composition. Standard errors clustered by 1920 state are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The broadest sample contains 4.88 million children and produces a 95 percent attendance interval that ranges from -0.043 to 0.016 . A wild-cluster Anderson–Rubin test does not reject zero ($p = 0.259$) but rejects a five-percentage-point benefit ($p < 0.001$). Broader family coverage therefore does not reveal a school attendance gain under the preferred age controls.

D.4 Completed education in 1940

Valid 1940 links are available for 51.6 percent of eligible boys and 14.8 percent of girls. We study the 451,232 sons with observed education, using the our primary 1920 treatment, instruments, and fixed effects. Outcomes are years of education, capped at 17, and completion of grades 8 and 12. We do not treat missing education codes as non-completion. Table 14 also reports 1920 school attendance within the same sample for robustness. We see that the first stage is strong but the educational attainment estimates are again imprecise.

E Predetermined-linkage selection

The linked-background (to 1910) sample contains 2,896,783 child records before the directional and nonmissing-background restrictions. We exclude 3,694 records because children in the same 1920

Table 14: Older-Sibling Work and Completed Education in 1940

Outcome	Linked sons	Reweightd
Attendance, 1920	0.007 (0.022)	0.023 (0.030)
Years of education, 1940	−0.257 (0.427)	−0.286 (0.424)
Completed grade 8	−0.052 (0.063)	−0.053 (0.060)
Completed grade 12	0.014 (0.048)	0.018 (0.044)
Observations	451,232	451,232
Families	405,642	405,642
States	48	48

Notes: The treatment is the share of strictly older siblings aged 10–16 in market work. All models apply state-by-focal-child-age and exact older-sibling-age-roster fixed effects. Focal children in this case are sons with unambiguous 1940 links and observed school attainment. Years of education are capped at 17. Grade 8 and grade 12 indicate completion. Standard errors clustered by 1920 state are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

family map to different 1910 origins. Compared with the unlinked full-count sample, linked children are more likely to come from a farm family (39.1 versus 27.3 percent) and somewhat more likely to attend school (90.1 versus 86.2 percent). They are also less likely to have an illiterate father (5.1 versus 13.3 percent).

We construct family-level calibration weights separately for the paternal-occupation and parental-literacy samples, calibrating to the full-count sample of younger children with at least one strictly older sibling. Iterative proportional fitting matches state, sampled-child ages, younger children, sex composition, urban and farm status, race, parental ages, literacy, immigrant status, and housing tenure. Every target category has observations that we can link on. We cap weights at the 99th percentile. Table 15 reports the effective sample sizes.

Table 15: Reweighting the Linked 1910 Sample

1910 family background	Unweighted	Reweighted
All linked families	0.008 (0.024)	0.014 (0.026)
Low paternal occupation score	0.012 (0.034)	0.024 (0.039)
Higher paternal occupation score	0.007 (0.018)	0.003 (0.018)
At least one illiterate parent	0.001 (0.050)	0.016 (0.050)
No illiterate parent	−0.001 (0.020)	−0.002 (0.019)
<i>Calibration diagnostics</i>		
Paternal-occupation sample: effective observations	1,159,371 of 1,217,941	
Parental-literacy sample: effective observations	1,200,413 of 1,259,508	

Notes: Entries are two-stage least-squares coefficients for younger-child attendance, with standard errors in parentheses. Reweighted estimates calibrate linked families to the eligible full-count 1920 sample on state, sampled-child age composition, younger children, sex composition, urban and farm status, race, parental ages, literacy, immigrant status, and housing tenure. Weights are capped at the 99th percentile. Reweighting addresses observed differences, not unobserved determinants of census linkage. Standard errors are clustered by state. Stars use a t distribution with the number of states minus one degrees of freedom. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The reweighted school attendance estimates remain small and imprecise, though we note that we cannot correct for unobserved correlates of linkage probability or recover families that had dissolved before 1920.

F Siblings near the legal work threshold

For younger children aged 14 or 15, all of their older siblings are already eligible to work under both of the most common state laws (with age thresholds of 12 and 14). Ultimately, we cannot test the effect on these younger children but, as a similar check, we can focus on children aged 10–11 who are near the threshold and whose older siblings are aged 12–14. This exercise asks whether the work status of siblings who are somewhat near the legal threshold affects younger children’s school attendance.

Table 16 retains all of the school attendance outcomes from the primary exercise and we see that, again, the estimates are small and imprecise. The low-paternal-standing and farm-family estimates are negative, but neither differs significantly from its comparison group (the differences are −0.255

Table 16: School Attendance with Siblings Near the Legal Work Threshold

Sample and potential earner	School attendance
<i>Panel A. Full-count comparisons</i>	
Older brother, children aged 10–11	−0.064 (0.061)
Older sister, children aged 10–11	−0.126 (0.171)
Older brother, children aged 10–12	−0.046 (0.066)
<i>Panel B. Family-background comparisons</i>	
At least one illiterate parent	−0.180 (0.180)
Low paternal occupational standing	−0.330* (0.167)
Higher paternal occupational standing	−0.076 (0.150)
Father in farm work in 1910	−0.343 (0.207)
Father outside farm work in 1910	−0.180* (0.093)
Low paternal standing, older sister	−0.009 (0.143)

Notes: All entries use school attendance as the outcome. Focal children are aged 10–11 unless otherwise noted. Potential earners are strictly older siblings aged 12–14. Models use the age-12 and age-14 laws, law-specific exposure instruments, state-by-focal-child-age-by-sex effects, and exact older-sibling age-by-sex roster effects. Paternal standing and farm status are measured in 1910. Parental illiteracy is measured in 1920. Standard errors are clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(0.172) and −0.163 (0.249), respectively).

G Data and Code Availability

The deposited replication archive for [Manacorda \(2006\)](#) contains the original one-percent extract, child labor regime indicator files, and archived programs. Our code, sample crosswalks, estimation scripts, and output tables will be deposited with this paper. The full-count census and Census Tree links are used under their respective data-use conditions.